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(54) **WORKSPACE-BASED FIXED PASS-THROUGH MONITORING SYSTEM AND METHOD FOR HARDWARE DEVICES USING A BASEBOARD MANAGEMENT CONTROLLER (BMC)**

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H04L 9/40 (2022.01)
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CPC H04L 63/1466; H04L 63/0218; G06F 11/141; G06F 2201/86
See application file for complete search history.

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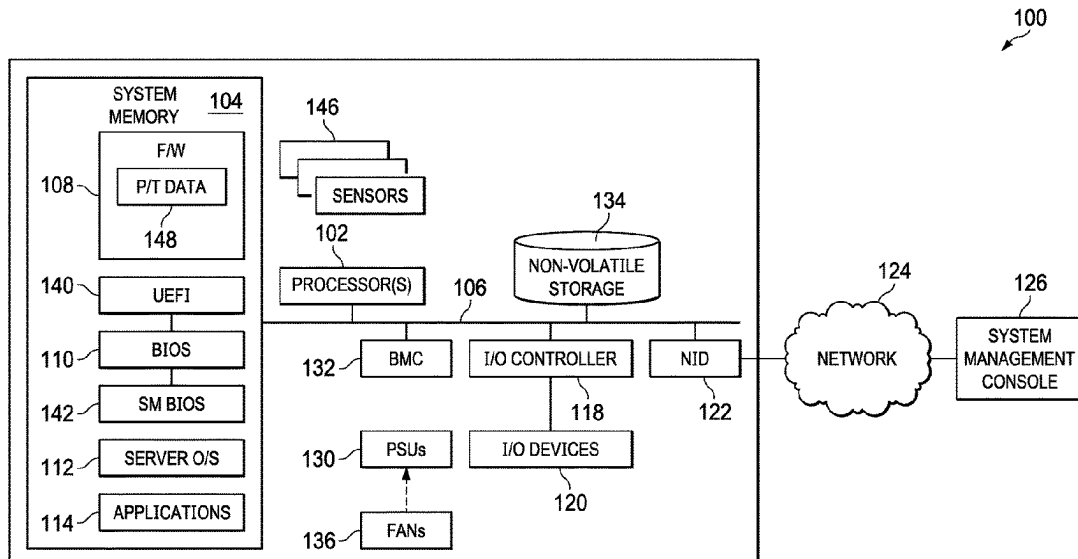
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(57) **ABSTRACT**

An Information Handling System (IHS) includes multiple hardware devices, and a baseboard Management Controller (BMC) in communication with the plurality of hardware devices. The BMC includes executable instructions for monitoring the operating characteristics a hardware device that is operating in a fixed pass-through configuration with a workspace in which the workspace has been instantiated by a workspace orchestration service executed on the IHS. The executable instructions may determine that the operating characteristics are indicative of a security breach of the fixed pass-through configuration, and as such, may perform an operation to quarantine the one hardware device when the fixed pass-through configuration is determined to possess the security breach.

12 Claims, 8 Drawing Sheets



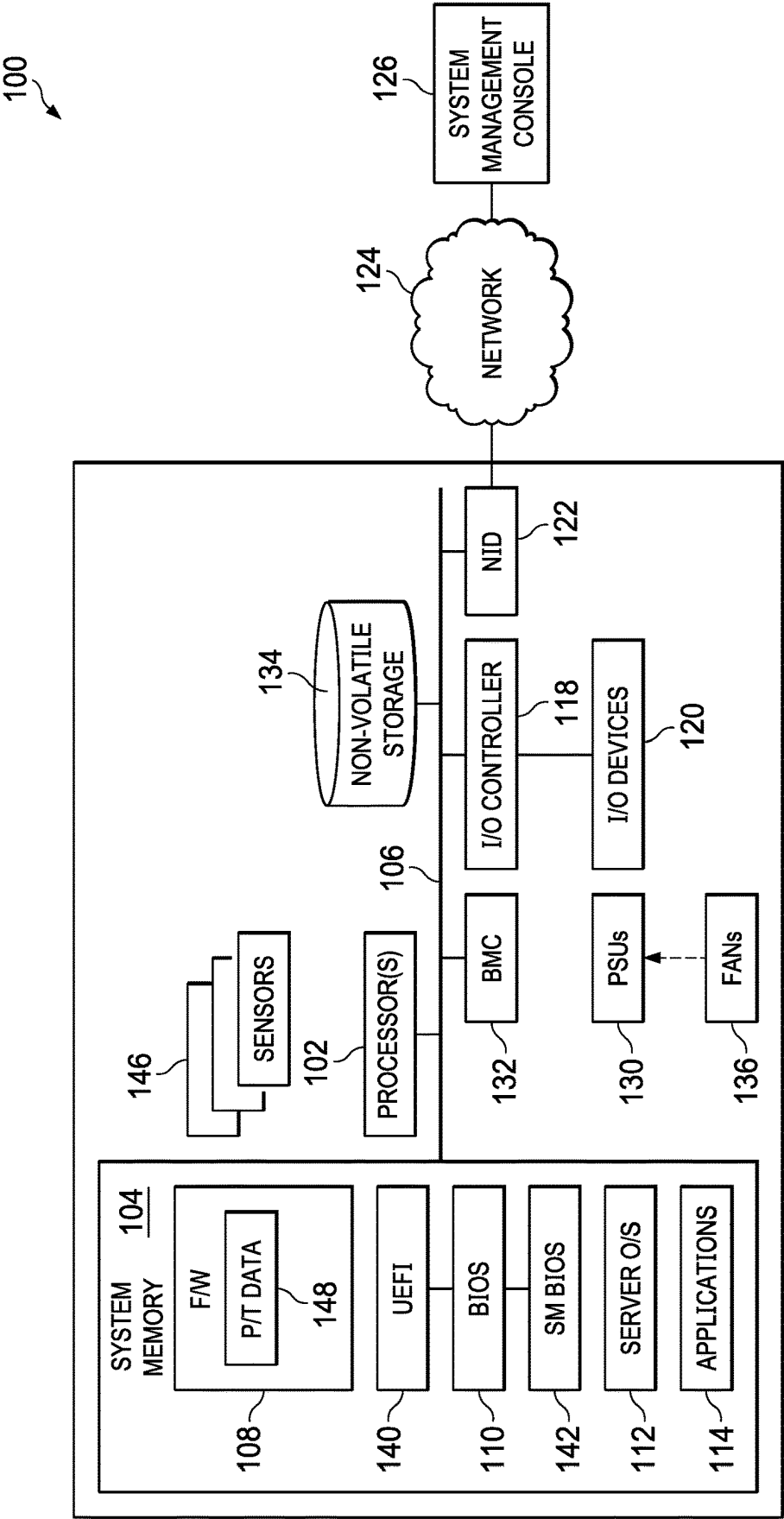


FIG. 1

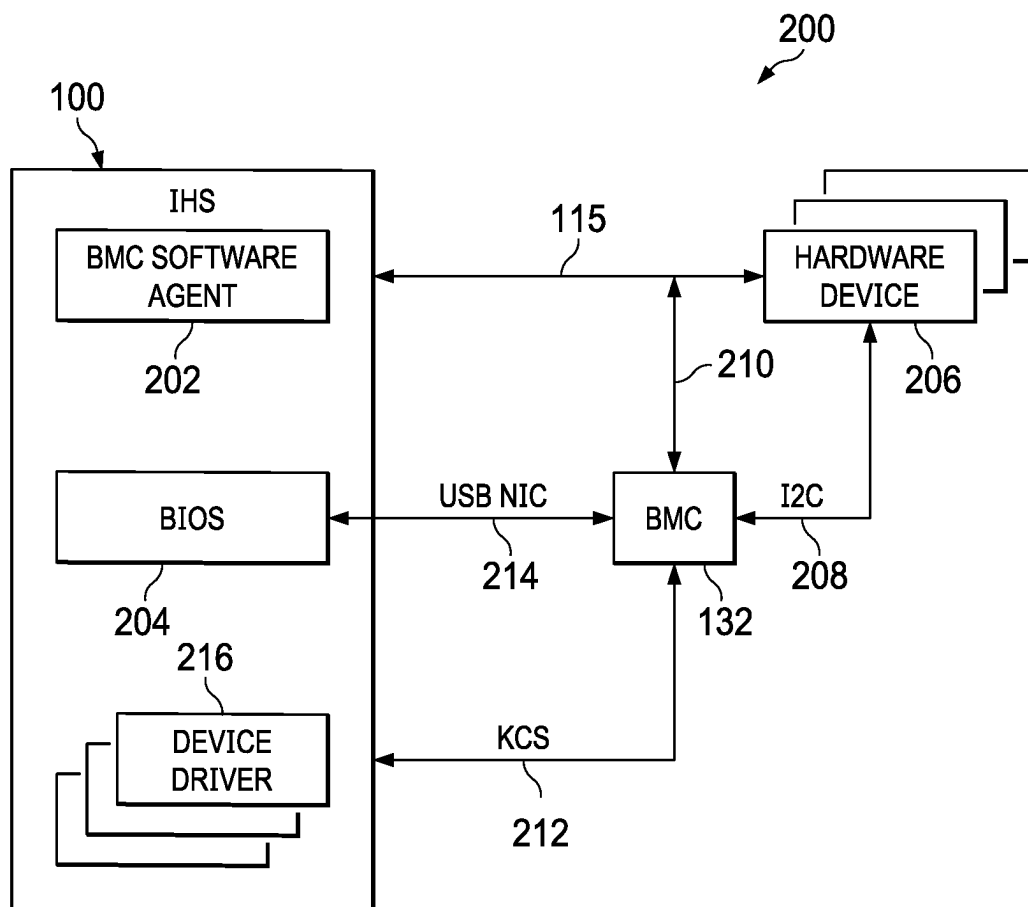
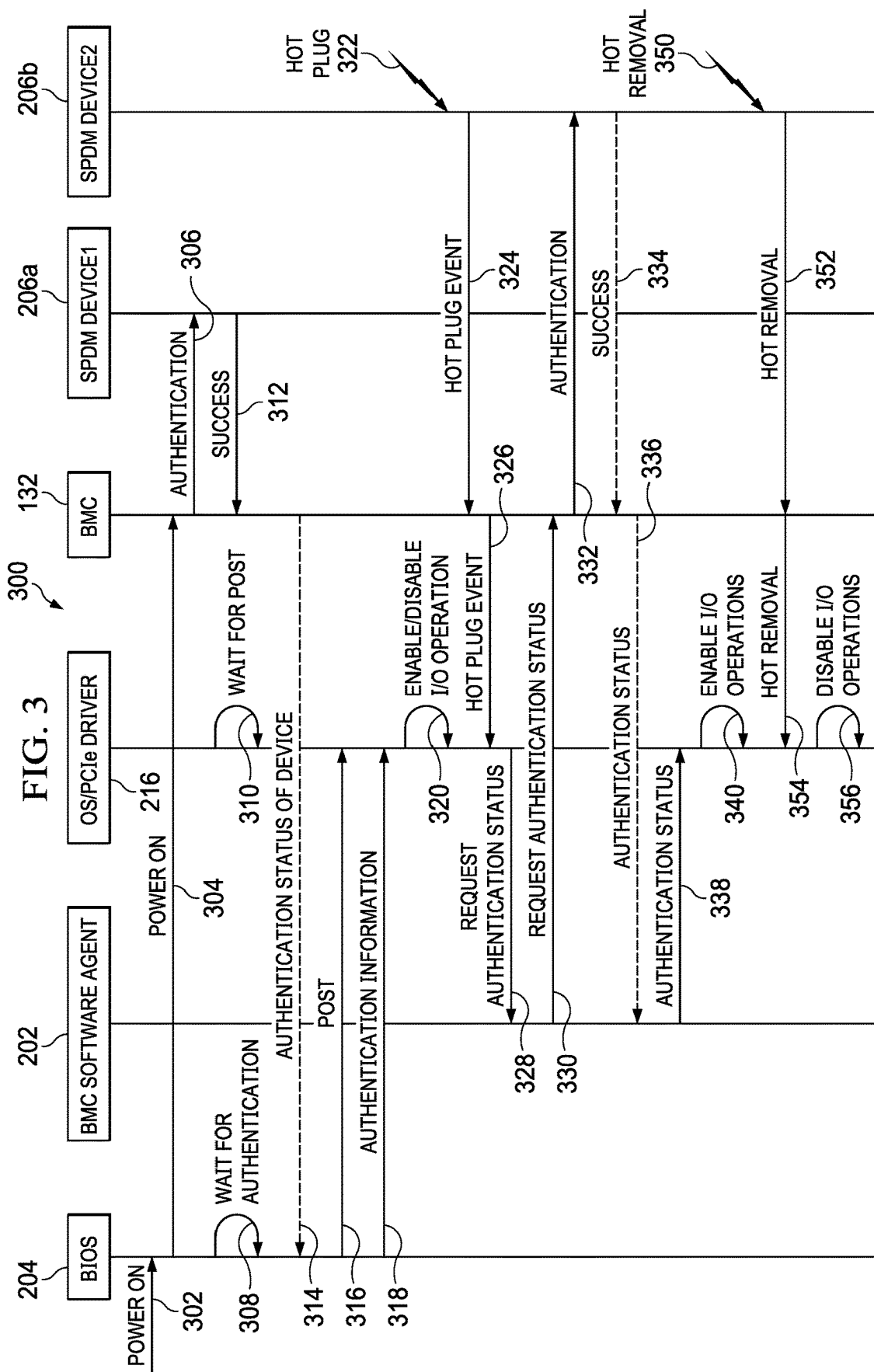
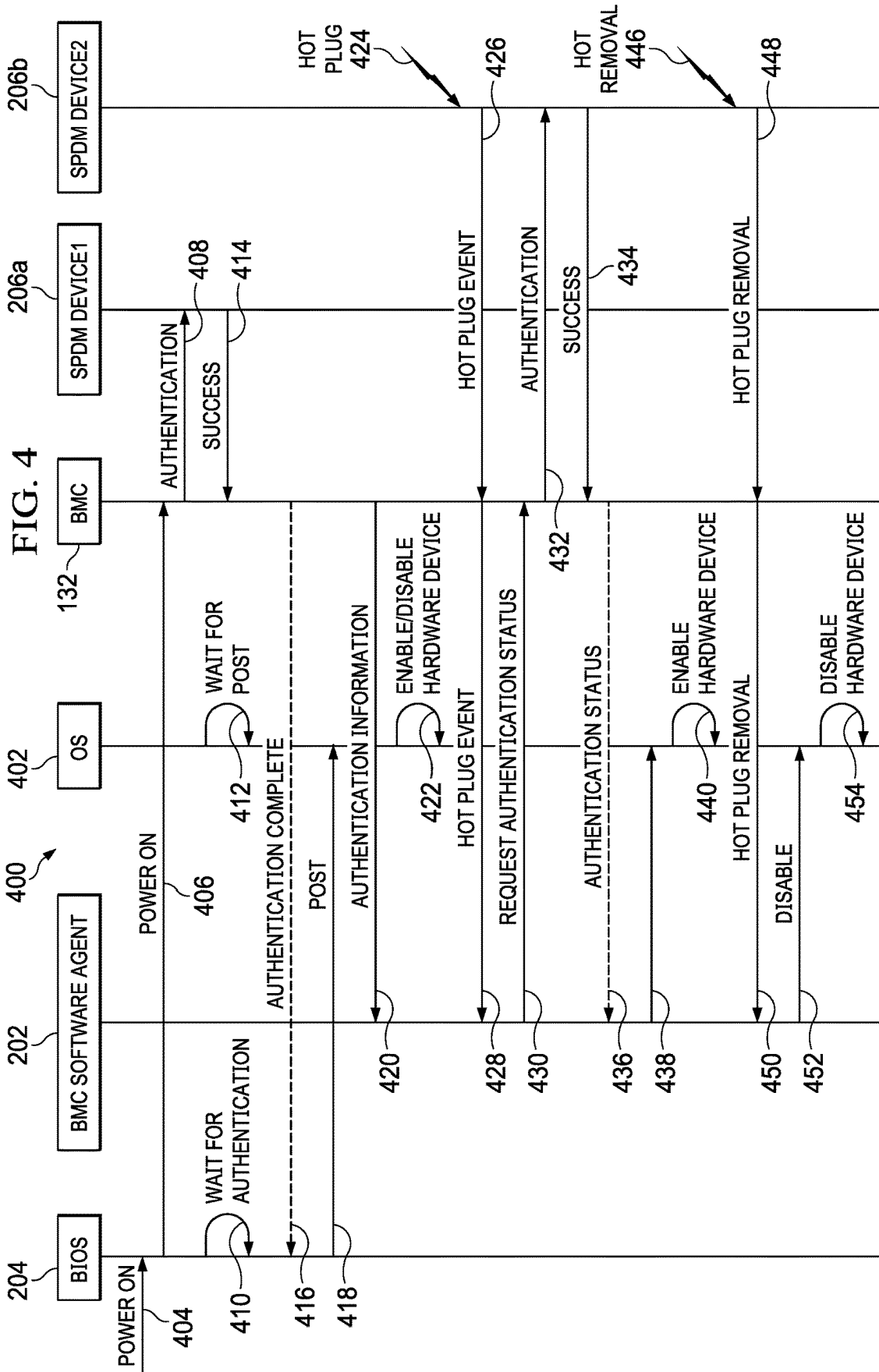


FIG. 2

FIG. 3





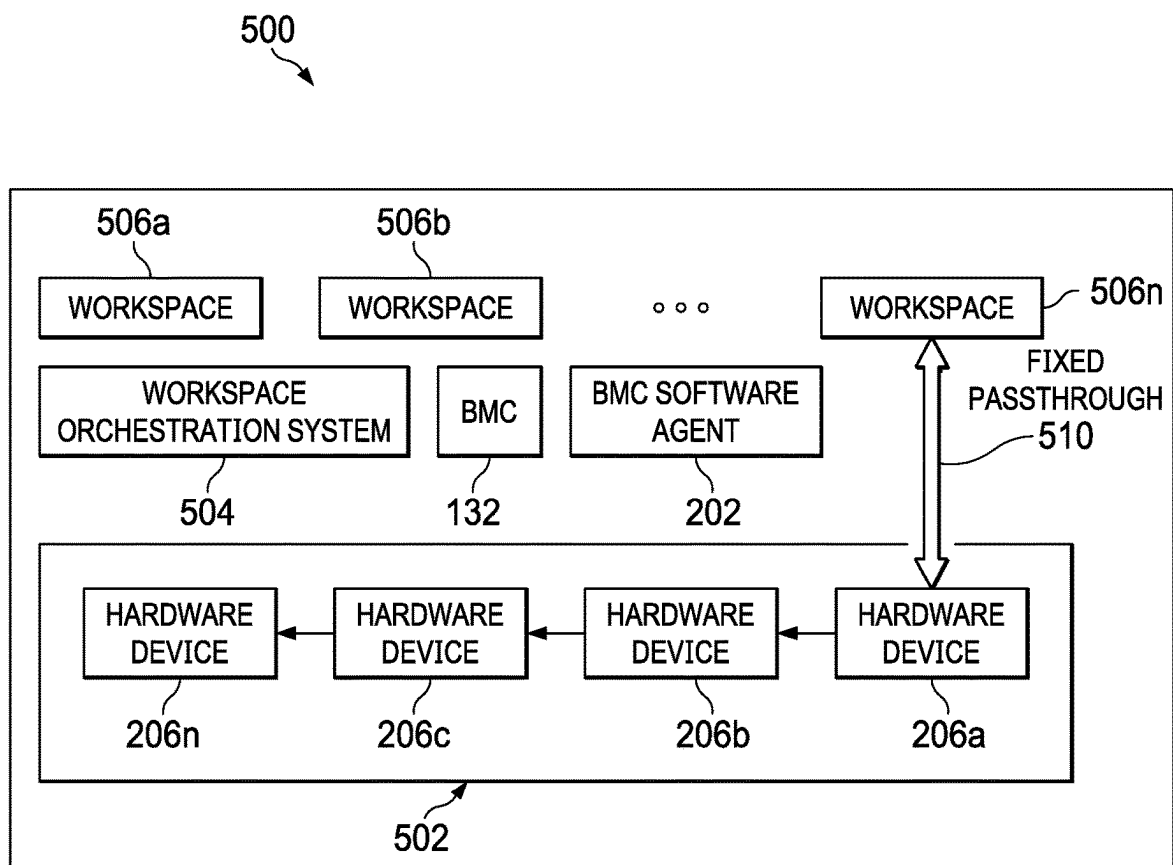


FIG. 5

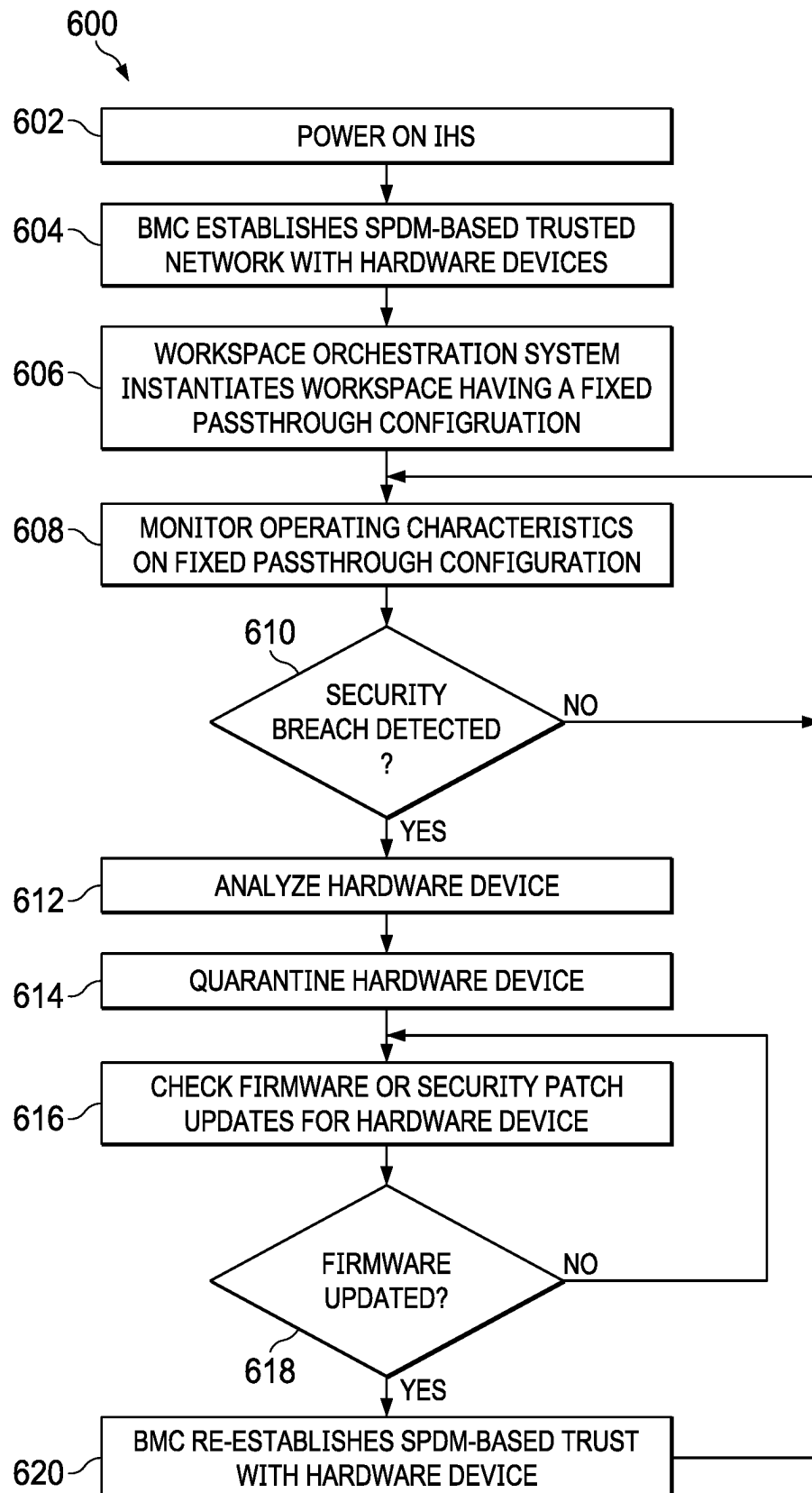


FIG. 6

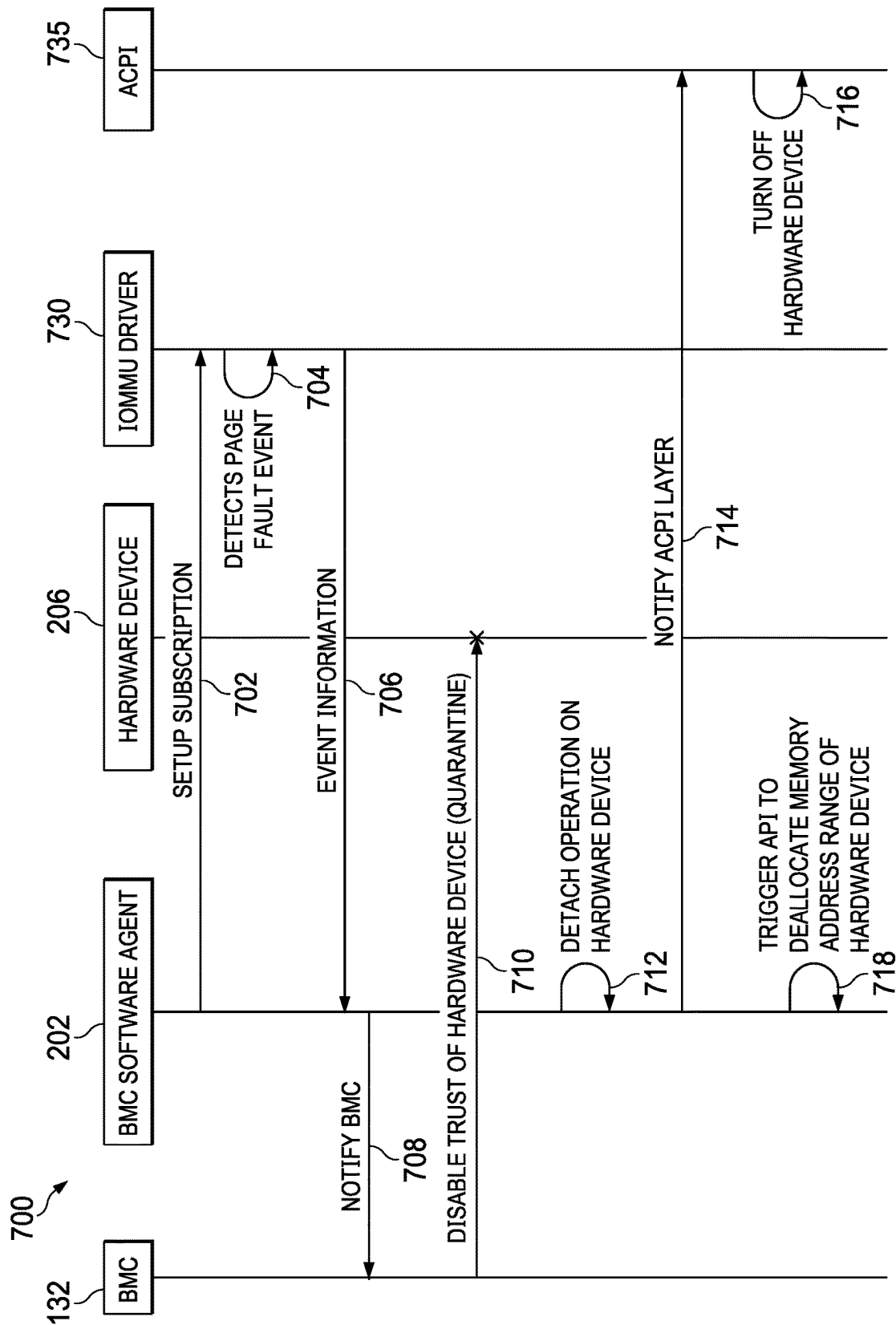


FIG. 7

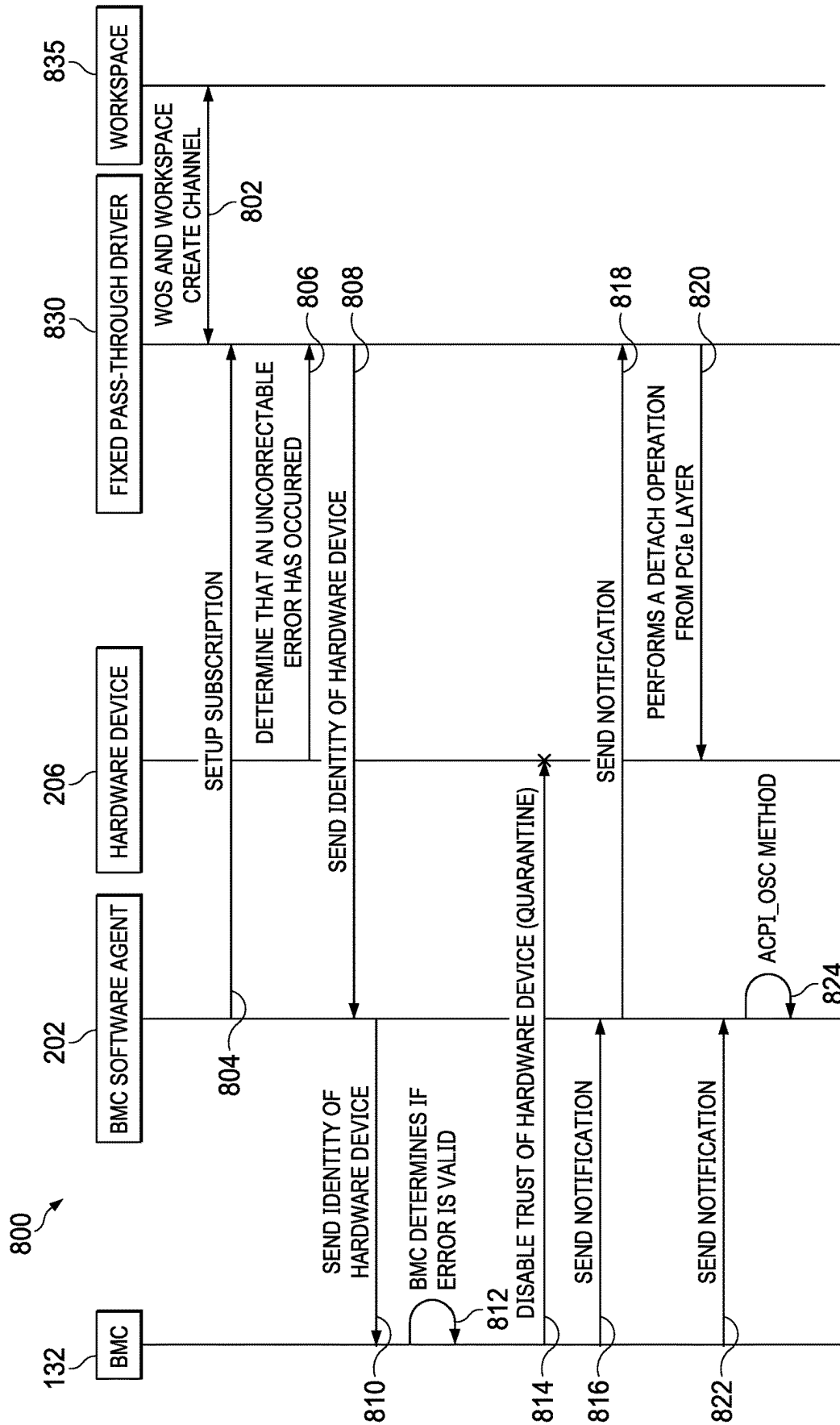


FIG. 8

1

**WORKSPACE-BASED FIXED
PASS-THROUGH MONITORING SYSTEM
AND METHOD FOR HARDWARE DEVICES
USING A BASEBOARD MANAGEMENT
CONTROLLER (BMC)**

CROSS REFERENCE TO RELATED
APPLICATIONS

This patent application claims priority to co-pending, commonly assigned Indian Patent Application No. 202111030705, filed Jul. 8, 2021 and entitled “Workspace-Based Fixed Pass-Through Monitoring System and Method for Hardware Devices Using a Baseboard Management Controller (BMC);” the entire contents of which are incorporated by reference herein.

FIELD

This disclosure relates generally to Information Handling Systems (IHSs), and more specifically, to a workspace-based fixed pass-through monitoring system and method for hardware devices using a baseboard management controller (BMC).

BACKGROUND

An Information Handling System (IHS) generally refers to any computing system that processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes. Modern day IHS administrative management is often provided via baseboard management controllers (BMCs). The baseboard management controller (BMC) generally includes a specialized microcontroller embedded in the IHS, and may provide an interface between system-management software and platform hardware. Different types of sensors built into the IHS report to the BMC on parameters such as temperature, cooling fan speeds, power status, operating system (O/S) status, and the like. The BMC monitors the sensors and can send alerts to a system administrator via the network if any of the parameters do not stay within pre-set limits, indicating a potential failure of the system. The administrator can also remotely communicate with the BMC to take certain corrective actions, such as resetting or power cycling the system to get a hung O/S running again. These abilities can often save on the total cost of ownership of an IHS, particularly when implemented in large clusters, such as server farms.

Many IHSs provide users with capabilities for accessing, creating, and manipulating data. IHSs often implement a variety of security protocols in order to protect this data during such operations. A known technique for securing access to protected data that is accessed via an IHS is to segregate the protected data within an isolated software environment in the form of a workspace that operates on the IHS, where such isolated software environments may be referred to by various names, such as virtual machines (VMs), containers, dockers, etc. Various types of such workspaces are isolated by providing varying degrees of abstraction from the underlying hardware and from the operating system of the IHS. The workspace is capable of instantiating multiple instances of guest operating systems (OSs) on the same “bare-metal” IHS using a supervisory software layer commonly referred to as a hypervisor, which may be interchangeably referred to as a host.

To enhance the performance of VMs, a fixed pass-through feature was implemented that directly assigned certain hard-

2

ware devices (e.g., graphics processors, network interface cards (NICs), storage controllers, etc.) to a VM. For example, processor vendors added hardware support for device virtualization, such as VT-d provided by INTEL and I/O management unit (IOMMU) provided by AMD. This feature allows a single guest OS instance to exclusively own a hardware device. Moreover, other VMs concurrently running on the IHS would be inhibited from using any hardware device that was currently operating in a fixed pass-through configuration. This configuration allows a single guest OS to achieve close to native performance for the hardware device. PCIe pass-through is one example of a fixed pass-through configuration that has been implemented for use on PCIe system buses.

SUMMARY

An Information Handling System (IHS) includes multiple hardware devices, and a baseboard Management Controller (BMC) in communication with the plurality of hardware devices. The BMC includes executable instructions for monitoring the operating characteristics of a hardware device that is operating in a fixed pass-through configuration with a workspace in which the workspace has been instantiated by a workspace orchestration service executed on the IHS. The executable instructions may determine that the operating characteristics are indicative of a security breach of the fixed pass-through configuration, and as such, may perform an operation to quarantine the one hardware device when the fixed pass-through configuration is determined to possess the security breach.

According to another embodiment, a method includes the steps of monitoring one or more operating characteristics of one or more hardware devices that is operating in a fixed pass-through configuration with a workspace using a Baseboard Management Controller (BMC). The hardware devices are configured in an Information Handling System (IHS). Additionally, the workspace has been instantiated by a workspace orchestration service executed on the IHS. The method further includes the steps of determining that the operating characteristics are indicative of a security breach of the fixed pass-through configuration, and performing an operation to quarantine the one hardware device when the fixed pass-through configuration is determined to possess the security breach.

According to yet another embodiment, an Information Handling System (IHS) includes first and second hardware devices in which the first hardware device includes executable instructions to monitor the operating characteristics of the second hardware device that is operating in a fixed pass-through configuration with a workspace. The workspace has been instantiated by a workspace orchestration service executed on the IHS. The instructions may also determine that the operating characteristics are indicative of a security breach of the fixed pass-through configuration, and as such perform an operation to quarantine the second hardware device when the fixed pass-through configuration is determined to possess the security breach.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention(s) is/are illustrated by way of example and is/are not limited by the accompanying figures, in which like references indicate similar elements. Elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale.

3

FIG. 1 shows an example IHS configured to implement a workspace-based fixed pass-through monitoring system and method for hardware devices using a baseboard management controller (BMC).

FIG. 2 is a diagram illustrating several components of an example BMC along with those of an associated IHS showing how those components may signal one another for implementing the BMC-based hardware device authentication system according to one embodiment of the present disclosure.

FIG. 3 is a workflow diagram describing certain steps of one embodiment of a hardware device authentication method according to one embodiment of the present disclosure.

FIG. 4 is a workflow diagram describing certain steps of another embodiment of a hardware device authentication method according to one embodiment of the present disclosure.

FIG. 5 illustrates an example IHS that may use a SPDMM-based trusted network to remediate security vulnerabilities caused by fixed pass-through configured in a virtualized environment according to one embodiment of the present disclosure.

FIG. 6 illustrates an example fixed pass-through monitoring method for hardware devices of an information handling system (IHS) using a baseboard management controller (BMC) according to one embodiment of the present disclosure.

FIG. 7 is a workflow diagram describing certain steps of another fixed pass-through monitoring and security breach remediation method using a software driver associated with the processor of the IHS according to one embodiment of the present disclosure.

FIG. 8 is a workflow diagram describing certain steps of yet another fixed pass-through monitoring and security breach remediation method by monitoring uncorrectable errors of the fixed pass-through configuration according to one embodiment of the present disclosure

DETAILED DESCRIPTION

Embodiments of the present disclosure provide a workspace-based fixed pass-through monitoring system and method for hardware devices of an information handling system (IHS) using a baseboard management controller (BMC). While fixed pass-through configurations may allow a hardware device to operate at near native performance, their use has engendered certain security threats due at least in part to their lack of transparency from other components whose purpose exists to ensure protection from those security threats. Embodiments of the present disclosure provide a solution to this problem, among others, using a system that continually monitors the activities of fixed pass-through configurations of workspaces to detect potential security breaches, such that when they are found, an affected hardware device may be quarantined until proper remediation techniques are employed to alleviate the security breach.

Certain IHSs may be configured with BMCs that are used to monitor, and in some cases manage computer hardware components of their respective IHSs. A BMC is normally programmed using a firmware stack that configures the BMC for performing out-of-band (e.g., external to a computer's operating system or BIOS) hardware management tasks. The BMC firmware can support industry-standard Specifications, such as the Intelligent Platform Management

4

Interface (IPMI) and Systems Management Architecture of Server Hardware (SMASH) for computer system administration.

Many IHS designs have been implemented with various types of system buses to enhance their functionality and extensibility. In general, a system bus provides an electrical interconnection scheme for connecting the hardware devices of an IHS, often using one or more expansion slots. The system bus combines the functions of a data bus to convey information, an address bus to determine where it should be sent, and a control bus to determine how the information is handled. Examples of such system buses may include those conforming to a Peripheral Component Interconnect (PCI) standard, a Peripheral Component Interconnect Express (PCIe) standard, or an accelerated graphics port (AGP) standard. Examples of hardware devices that may use these system buses include graphics processing cards, network interface cards, input/output (I/O) (e.g., USB) expansion port cards, Wi-Fi interface cards, storage controller cards, and the like.

These system buses have provided a relatively efficient technique for modifying or enhancing the functionality of an IHS by enabling the hardware devices to be easily added to, and removed from the IHS. Many of these hardware devices are provided with their own computer executable environment capable of performing certain algorithms. These algorithms, however, may include harmful (either malicious or inadvertent) algorithms that can damage or hinder the operation the IHS if not managed properly.

A fixed pass-through feature, which is now commonly available with many virtualized environments (e.g., workspace orchestration systems), has engendered certain security vulnerabilities. One example of such a fixed pass-through feature is a PCIe pass-through that may be utilized with PCIe system buses. A virtualized environment may include a hypervisor or virtual machine manager (VMM) that can instantiate multiple instances of virtual machines (VMs) (e.g., guest operating systems (OSs)) on the same "bare-metal" IHS. The fixed pass-through configuration allows a virtual machine (VM) instance to have direct access to a hardware device on the IHS. Direct device assignment (fixed pass-through) is such a mechanism, where the VMM assigns a device exclusively to one VM instead of sharing it with other virtual machines. While assigning a hardware device to be directly controlled by a VM improves its performance, this approach also introduces a wide range of security vulnerabilities.

Using fixed pass-through to a virtual machine often results in potential security vulnerabilities. For example, though hardware device is authenticated through certificate and firmware measurements, some flaw in the hardware device firmware or its associated driver can still make the hardware device vulnerable. The vulnerability can be triggered when buggy or malicious code, such as a device driver or a firmware is running in privileged mode on the VM (e.g., guest OS). Moreover, attackers that successfully exploit vulnerabilities in the host may also compromise the privacy and integrity of other VMs operating on the IHS. The flaw could be applicable to other device sharing modes such as SR-IOV, and the like. As will be described in detail herein below, embodiments of the present disclosure provide a solution to these problems, among others, by providing a workspace fixed pass-through monitoring system and method for hardware devices of an information handling system (IHS) using a baseboard management controller (BMC).

For purposes of this disclosure, an IHS may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an IHS may be a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., Personal Digital Assistant (PDA) or smart phone), server (e.g., blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. An IHS may include Random Access Memory (RAM), one or more processing resources such as a Central Processing Unit (CPU) or hardware or software control logic, Read-Only Memory (ROM), and/or other types of nonvolatile memory.

Additional components of an IHS may include one or more disk drives, one or more network ports for communicating with external devices as well as various I/O devices, such as a keyboard, a mouse, touchscreen, and/or a video display. As described, an IHS may also include one or more buses operable to transmit communications between the various hardware components. An example of an IHS is described in more detail below.

FIG. 1 shows an example of an IHS 100 configured to implement a workspace-based fixed pass-through monitoring system and method for hardware devices using a base-board management controller (BMC). In one embodiment, the IHS 100 may implement systems and methods for hardware device authentication using the BMC. Particularly, IHS 100 includes one or more processor(s) 102 coupled to system memory 104 via system bus which may include any suitable type. System memory 104 may include a plurality of software and/or firmware modules including firmware (F/W) 108, basic input/output system (BIOS) 110, operating system (O/S) 112, and/or application(s) 114. Software and/or firmware module(s) stored within system memory 104 may be loaded into processor(s) 102 and executed during operation of IHS 100.

F/W 108 may include a power/thermal profile data table 148 that is used to store power profile data and thermal profile data for certain hardware devices (e.g., processor(s) 102, system memory 104, non-volatile storage 134, NID 122, I/O controllers 118, etc.). System memory 104 may include a UEFI interface 140 and/or a SMBIOS interface 142 for accessing as well as updating the BIOS 110. In general, UEFI interface 140 provides a software interface between an operating system and BIOS 110. In many cases, UEFI interface 140 can support remote diagnostics and repair of computers, even with no operating system installed. SMBIOS interface 142 can be used to read management information produced by BIOS 110 of a IHS 100. This feature can eliminate the need for the operating system to probe hardware directly to discover what devices are present in the computer.

IHS 100 includes one or more input/output (I/O) controllers 118 which manages the operation of one or more connected input/output (I/O) device(s) 120, such as a keyboard, mouse, touch screen, microphone, a monitor or display device, a camera, a microphone, audio speaker(s) (not shown), an optical reader, a universal serial bus (USB), a card reader, Personal Computer Memory Card International Association (PCMCIA) slot, and/or a high-definition multimedia interface (HDMI), which may be included or coupled to IHS 100.

IHS 100 includes Network Interface Device (NID) 122. NID 122 enables IHS 100 to communicate and/or interface with other devices, services, and components that are located externally to IHS 100. These devices, services, and components, such as a system management console 126, can interface with IHS 100 via an external network, such as network 124, which may include a local area network, wide area network, personal area network, the Internet, etc.

IHS 100 further includes one or more power supply units (PSUs) 130. PSUs 130 are coupled to a BMC 132 via an I²C bus. BMC 132 enables remote operation control of PSUs 130 and other components within IHS 100. PSUs 130 power the hardware devices of IHS 100 (e.g., processor(s) 102, system memory 104, non-volatile storage 134, NID 122, I/O controllers 118, PSUs 130, etc.). To assist with maintaining temperatures within specifications, an active cooling system, such as one or more fans 136 may be utilized.

IHS 100 further includes one or more sensors 146. Sensors 146 may, for instance, include a thermal sensor that is in thermal communication with certain hardware devices that generate relatively large amounts of heat, such as processors 102 or PSUs 130. Sensors 146 may also include voltage sensors that communicate signals to BMC 132 associated with, for example, an electrical voltage or current at an input line of PSU 130, and/or an electrical voltage or current at an output line of PSU 130.

BMC 132 may be configured to provide out-of-band management facilities for IHS 100. Management operations may be performed by BMC 132 even if IHS 100 is powered off, or powered down to a standby state. BMC 132 may include a processor, memory, and an out-of-band network interface separate from and physically isolated from an in-band network interface of IHS 100, and/or other embedded resources.

In certain embodiments, BMC 132 may include or may be part of a Remote Access Controller (e.g., a DELL Remote Access Controller (DRAC) or an Integrated DRAC (iDRAC)). In other embodiments, BMC 132 may include or may be an integral part of a Chassis Management Controller (CMC). As a non-limiting example, the BMC 132 may be an integrated Dell Remote Access Controller (iDRAC) from Dell® that is embedded within a Dell PowerEdge™ server to provide functionality that helps information technology (IT) administrators deploy, update, monitor, and maintain servers remotely.

FIG. 2 is a diagram 200 illustrating several components of an example BMC 132 along with those of an associated IHS 100 showing how those components may signal one another for implementing the BMC-based hardware device authentication system according to one embodiment of the present disclosure. The IHS 100 is shown with a BMC software agent 202 and a basic input output system (BIOS) 204, and one or more device drivers 216 that are each associated with one of the hardware devices 206. The IHS 100 also includes one or more hardware devices 206, such as graphics processing cards, network interface cards, input/output (I/O) (e.g., USB) expansion port cards, Wi-Fi interface cards, storage controller cards, and the like. For example, the hardware devices 206 may be an I/O controller 118, a NID 122, or a storage controller configured to manage the non-volatile storage 134, such as described herein above. Each hardware device 206 communicates with the IHS 100 via system bus 106. In one embodiment, system bus 106 may include a Peripheral Component Interconnect Express (PCIe) bus.

According to certain embodiments of the present disclosure, the hardware device 206 may comprise any removable

device that communicates using the Security Protocol and Data Model (SPDM) protocol. Within this disclosure, a removable device may be any type that can be added to or removed from the IHS 100. In some cases, the hardware device 206 can be hot-plugged into the IHS 100. That is, the hardware device 206 can be added to, or removed from the IHS 100 while it is running.

The BMC 132 communicates with each of multiple hardware devices 206 via an i2c connection 208 such as described above. In other embodiments, the BMC 132 may communicate with each of the hardware devices 206 using other communication protocol, such as i3c SENSEWIRE connections, or serial peripheral interface (SPI) based connections. Additionally, the BMC 132 may also communicate with each of the hardware devices 206 through the system bus 106 using a Management Component Transport Protocol (MCTP) PCIe vendor defined message (VDM) channel 210.

The BMC 132 may also communicate with the IHS 100 through the PCIe VDM channel 210, a Keyboard Controller Style (KCS) channel 212, and/or a USB NIC channel 214. The KCS channel 212 is provided as part of an intelligent platform management interface (IPMI) specification. In general, the IPMI specification defines several system interfaces for local access from the BMC software agent 202 and BIOS 204 to the BMC 132 of which the KCS protocol provides an ideal connection that does not require authentication for its operation.

The BMC software agent 202 is a lightweight software service that is executed on the host IHS 100 to integrate certain operating system (OS) features with the BMC 132. The BMC software agent 202 provides OS-related information to the BMC 132, and may add capabilities such as LC log event replication into the OS log, WMI support (including storage), iDRAC SNMP alerts via OS, iDRAC hard reset and remote full Power Cycle. For example, the BMC software agent 202 may be an iDRAC Service Module (ISM) that is configured to operate with the integrated Dell remote access controller (iDRAC), which are both provided by the DELL CORPORATION.

As described above, some hardware devices, which can be physically added to and removed from IHSs can cause certain security threats if not handled properly. Many of these hardware devices are provided with their own computer executable environment capable of executing certain algorithms for providing their respective services. These algorithms, however, may include harmful (either malicious or inadvertent) code that can damage or hamper the operation the IHS. The hardware device authentication system uses an authentication scheme for safely entering the hardware devices into service for their IHSs while eliminating or reducing any potential security breaches that could be caused thereby.

During booting of the BMC 132, the hardware devices 206 are authenticated and a secure channel is established. Once authenticated, the hardware devices 206 can be trusted and I/O operations can be performed by the IHS 100. Nevertheless, certain hardware devices 206 may be hot plug capable (e.g., inserted while the IHS is running), thus yielding a security leak in which malicious code running on the hardware device may be able to access the system bus 106. Additionally, OS resource allocation and utilization by certain hardware devices 206 may start earlier than the BMC 132 (e.g., DELL's iDRAC) has a chance to authenticate the device using its sideband interface. As will be described in detail herein below, embodiments of the present disclosure cause the BIOS 204 power on self-test (POST) procedure to

wait until some, most, or all hardware devices 206 are authenticated prior to allowing the POST procedure to complete.

Although FIG. 2 describes one example of an architecture that may be configured to provide a hardware device authentication system, the features of the disclosed architecture may be embodied in other specific forms without deviating from the spirit and scope of the present disclosure. For example, the IHS/BMC architecture as shown may include additional, fewer, or different components than those operations as described in the present example. As another example, while the IHS 100 is shown with those components described above for purposes of brevity and clarity of discussion, it should be appreciated that IHS 100 may be configured with additional, fewer, or different components without departing from the spirit and scope of the present disclosure.

FIG. 3 is a workflow diagram describing certain steps of one embodiment of a hardware device authentication method 300 according to one embodiment of the present disclosure. As will be described in detail herein below, the method 300 may be provided for authenticating hardware devices 206 prior to allowing them to access the processes (e.g., I/O operations) of the host IHS 100. Initially, the IHS 100 is configured with a BMC 132 and a hardware device 206a that is capable of communicating using the SPDM protocol. Although only two hardware devices 206a, 206b are shown and described in the method 300, it should be understood that the method 300 can be practiced with any quantity of hardware devices 206, such as three or more SPDM enabled hardware devices 206.

At step 302, the IHS 100 is powered on. The IHS 100 may be powered on manually by a user, or remotely, such as a remote command received through the BMC 132. At step 304, the BIOS 204 issues a power ON signal to the BMC 132, which in turn, causes the BMC 132 to commence its authentication procedure with each of the hardware devices 206a in the IHS 100 at step 306.

Immediately after sending the power ON signal to the BMC 132, the BIOS 204 waits for authentication by the BMC 132 for each hardware device 206a at step 308. That is, the BIOS 204 may concurrently perform its POST in the normal manner, but is not allowed to issue the POST complete message until it receives an authentication complete message from the BMC 132. Additionally, the device driver 216 associated with the hardware device 206a waits for the POST complete message prior to commencing its operation at step 310.

At some point in time when the authentication procedure started for the hardware device 206a at step 306 is completed, the hardware device 206a issues a success message to the BMC 132 at step 312. For example, the success message issued by the hardware device 206a may include a response message to a challenge message included in the message at step 306. In one embodiment, the authentication procedure may be conducted according to the SPDM protocol. Additionally, although the method 300 only describes the authentication of one hardware device 206a, it should be appreciated that the authentication procedure may be performed for each of multiple hardware devices 206a configured in the IHS 100.

At step 312, after the BMC 132 has performed the authentication procedure for the hardware device 206a, it transmits the authentication status of the hardware device 206a to the BIOS 204 at step 314. Thereafter at steps 316 and 318, the BIOS 204 issues the POST complete message to the device driver 216 along with its associated hardware

device **206a** authentication status, respectively. In response, the device driver **216** either inhibits or allows operation of its respective hardware device **204** based upon the authentication status at step **320**. That is, if a particular device driver **216** receives a successful authentication status, it would then enable its associated hardware device **206a** for operation with the IHS **100**. On the other hand, if a particular device driver **216** receives a failed authentication status, it would then inhibit its associated hardware device **206a** from operation with the IHS **100**. In one embodiment, the BIOS **204** and/or other entity, such as the system bus controller may inhibit operation of any unused slots. This action may be particularly useful in cases where a hardware device **206a** is hot-plugged into the unused slot in which operation of the hot-plugged hardware device **206a** may be authenticated prior to being allowed to access the services of the host IHS **100**. Such may be the case as described below with reference to steps **322** through **340**.

At step **322**, another hardware device **206b** is hot-plugged into a slot of the IHS **100**. While the hardware device **206b** is being described herein as being hot-plugged into a slot, it should be understood that the device may be any mechanism or connector that allows the hardware device **206b** to be interconnected to the IHS **100**. Furthermore, it is contemplated that, rather than a slot, the hardware device **206b** may be interconnected to the IHS **100** using a wireless connection, such as Bluetooth connection, that interconnects a SPDMM capable hardware device **206b** to the IHS **100**. The BMC **132** is then notified of the hot-plug event at step **324**, and forwards the notification to the device driver **216** associated with that hardware device **206b** at step **326**. At step **328**, the device driver **216** then sends an authentication status request to the BMC software agent **202**, which in turn, forwards the authentication request to the BMC **132** at step **330**.

At step **332**, the BMC **132** commences its authentication procedure with the hardware device **206b**. At some point in time when the authentication procedure is completed, the hardware device **206b** issues a success message to the BMC **132** at step **334**. The success message may be, for example, a successful response message to a previous challenge message presented by the BMC **132** at step **332**. Thereafter, the BMC **132** sends the authentication status to the BMC software agent **202** at step **336**, which in turn, forwards the authentication status to the device driver **216** associated with that hardware device **206b** at step **338**. In response, the device driver **216** either inhibits or allows operation of its respective hardware device **206b** based upon the authentication status at step **340**. That is, if a particular device driver **216** receives a successful authentication status, it would then enable its associated hardware device **206a** for operation with the IHS **100**; otherwise, it inhibits its associated hardware device **206b** from operation with the IHS **100**.

Steps **350** through **356**, generally describe a sequence of actions that may be performed to disable operation (e.g., removal) of a slot or other interconnection mechanism when the hardware device **206b** is hot removed from the IHS **100**. Within this disclosure, hot removal generally refers to the act of removing the hardware device **206b** while the IHS **100** is currently running. At step **350**, the hardware device **206b** is hot removed from the IHS **100**. The BMC **132** is notified of this event at step **352**, and in turn, sends a hot removal message to the device driver **216** associated with the hardware device **206b** at step **354**. Throughout this disclosure, the terms 'signal' and 'message' may be used interchangeably to mean any informational technique that is transmitted from one component of the system to another. When the

device driver **216** receives the hot removal message, it disables operation of the slot at step **356**.

As can be clearly seen from the description above, the BIOS POST procedure waits for the authentication complete message from the BMC **132**. Additionally, the BMC **132** authenticates some, most, or all the SPDMM capable hardware devices **206** and passes the authentication information to the BIOS **204** via IPMI command over a suitable interface, such as a KCS interface or a memory block shared between the BMC **132** and BIOS **204**. The authentication status is then made available to the device driver **216**. The device driver **216** can then enable and/or disable the I/O operations for the slot. When the hardware device **206** is hot plugged, the device driver **216** can request the authentication status from BMC software agent **202**. The BMC software agent **202** may in turn, request the authentication status from the BMC **132**. Once the authentication credentials are verified successfully, the BMC software agent **202** may then pass the information to the device driver **216**. The device driver **216** may then enable or disable the functionality (e.g., I/O operations) of the hardware device **206** based on the authentication status. When the hardware device **206** is removed, the device driver **216** shall disable the I/O operation for the slot. Nevertheless, when use of the workspace migration method **300** is no longer needed or desired, the process ends.

Although FIG. **3** describes an example method that may be performed for managing event messages, the features of the method **300** may be embodied in other specific forms without deviating from the spirit and scope of the present disclosure. For example, the method **300** may perform additional, fewer, or different operations than those described in the present examples. As another example, certain steps of the method **300** may be performed by other components in the IHS **100** other than those described above.

FIG. **4** is a workflow diagram describing certain steps of another embodiment of a hardware device authentication method **400** according to one embodiment of the present disclosure. As will be described in detail herein below, the method **400** may be provided for authenticating hardware devices **206** prior to allowing them to function, such as performing I/O operations with the host IHS **100**. Initially, the IHS **100** is configured with a BMC **132** and a hardware device **206a** that is capable of communicating using the SPDMM protocol. Although only two hardware devices **206a**, **206b** are shown and described in the method **400**, it should be understood that the method **400** can be practiced with any quantity of hardware devices **206**, such as three or more SPDMM enabled hardware devices **206**.

At step **404**, the IHS **100** is powered on. The IHS **100** may be powered on manually by a user, or remotely, such as a remote command received through the BMC **132**. At step **406**, the BIOS **204** issues a power ON signal to the BMC **132**, which in turn, causes the BMC **132** to commence its authentication procedure with the hardware device **206a** in the IHS **100** at step **408**.

Immediately after sending the power ON signal to the BMC **132**, the BIOS **204** waits for authentication by the BMC **132** for the hardware device **206a** at step **410**. In other embodiments in which the IHS **100** includes multiple hardware devices, the BIOS **204** may wait until the authentication for all the hardware devices are complete. That is, the BIOS **204** may concurrently perform its POST in the normal manner, but is not allowed to issue the POST complete message until it receives an authentication complete message from the BMC **132**. Additionally, the operating system

11

402 of the IHS 100 waits for the POST complete message prior to commencing its operation at step 412.

At some point in time when the authentication procedure started for the hardware device 206a at step 408 is completed, the hardware device 206a issues a success message to the BMC 132 at step 414. For example, the success message issued by the hardware device 206a may include a response message to a challenge message included in the message at step 408. In one embodiment, the authentication procedure may be conducted according to the SPDm protocol.

Although the method 400 only describes the authentication of one hardware device 206a, it should be appreciated that the authentication procedure may be performed for each of multiple hardware devices 206a configured in the IHS 100. Additionally in certain embodiments, a subset of all the SPDm capable hardware devices 206a may be designated to perform an authentication procedure with the BMC 132. For example, the BIOS 204 may store information for enforcing the authentication procedure for those hardware devices 206a that are provided by a certain vendor. As another example, the BIOS 204 may store information for not requiring certain hardware devices 206a to undergo the authentication procedure, such as those hardware devices 206 having a level of trust such that authentication is not warranted. Yet another example, the BIOS 204 may store information about those hardware devices 206 that have recently been upgraded with new firmware, and enforce the authentication procedure for only those hardware devices 206 that have recently been upgraded.

At step 414, after the BMC 132 has performed the authentication procedure for the hardware device 206a in the IHS 100, it transmits the authentication status of the hardware device 206a to the BIOS 204 at step 416. Thereafter at steps 418, the BIOS 204 issues the POST complete message to the operating system 402 along with its associated hardware device 206a authentication status. In response, the operating system 402 either inhibits or allows operation of its respective hardware device 206a based upon the authentication status at step 422. That is, if a particular operating system 402 receives a successful authentication status, it would then enable its associated hardware device 206a for operation with the IHS 100. On the other hand, if the operating system 402 receives a failed authentication status, it would then inhibit its associated hardware device 206a from operation with the IHS 100.

Additionally, the BMC 132 sends the authentication status information to the BMC software agent 202 at step 420. This action may be particularly useful in cases where a hardware device 206a is hot-plugged into the unused slot in which operation of the hot-plugged hardware device 206a may be authenticated by the BMC software agent 202, thus alleviating any intervention that would otherwise be required by the BIOS 204 and/or operating system 402. Such may be the case as described below with reference to steps 424 through 440.

At step 424, another hardware device 206b is hot-plugged into a slot of the IHS 100. The BMC 132 is then notified of the hot-plug event at step 426, and forwards the notification to the BMC software agent 202 at step 428. At step 430, the BMC software agent 202 then sends an authentication status request to the BMC 132, which in turn, commences an authentication procedure with the newly connected hardware device 206b at step 432. At some point in time when the authentication procedure is completed, the hardware device 206b issues a success message to the BMC 132 at step 434. The success message may be, for example, a

12

successful response message to a previous challenge message presented by the BMC 132 at step 432.

Thereafter, the BMC 132 sends the authentication status to the BMC software agent 202 at step 436, which in turn, forwards the authentication status to the operating system 402 at step 438. In response, the operating system 402 either inhibits or allows operation of the hardware device 206b based upon the authentication status at step 440. That is, if the operating system 402 receives a successful authentication status, it would then enable its associated hardware device 206b for operation with the IHS 100; otherwise, it inhibits the hardware device 206b from operation with the IHS 100.

Steps 446 through 454, generally describe a sequence of actions that may be performed to disable operation of a slot or other interconnection mechanism when the hardware device 206b is hot removed from the IHS 100. At step 446, the hardware device 206b is hot removed from the IHS 100. The BMC 132 is notified of this event at step 448, and in turn, sends a hot removal message to the BMC software agent 202 at step 450. Thereafter at step 452, the BMC software agent 202 notifies the operating system 402 of the hot removal action, such that the operating system 402 can disable the slot at step 454.

As can be clearly seen from the description above, the BIOS POST procedure waits for the authentication complete message from the BMC 132. Additionally, the BMC 132 authenticates some, most, or all the SPDm capable hardware devices 206 and informs the BIOS 204, which waits until it receives the message to issue a POST complete message to the IHS 100. Upon POST complete, the BMC 132 passes the authentication information to BMC software agent 202. Based on the authentication status, the BMC software agent 202 disables the device driver 216 for the slot of any un-authenticated hardware device 206 in addition to any empty slots. Additionally, the BMC software agent 202 may receive hot plug events from BMC 132, and it may disable/enable the driver for the slot based on the SPDm authentication status received from BMC 132. The BMC software agent 202 may disable the device driver 216 for the slot upon receiving hot removal notification from the BMC 132. Additionally, the BMC software agent 202 may shall enable/disable the slots if the host operating system 402 supports such functionality. Nevertheless, when use of the workspace migration method 400 is no longer needed or desired, the process ends.

Although FIG. 4 describes an example method that may be performed for managing event messages, the features of the method 400 may be embodied in other specific forms without deviating from the spirit and scope of the present disclosure. For example, the method 400 may perform additional, fewer, or different operations than those described in the present examples. As another example, certain steps of the method 400 may be performed by other components in the IHS 100 other than those described above.

FIG. 5 illustrates an example IHS 500 that may use a SPDm-based trusted network 502 to remediate security vulnerabilities caused by fixed pass-through configured in a virtualized environment according to one embodiment of the present disclosure. The SPDm-based trusted network 502 may be one in which the hardware devices 206 A-N (or individually referred to as hardware device 206) have been authenticated using the BMC-based hardware device authentication system and method as described above with reference to FIGS. 1-4. The hardware devices 206 A-N may be any SPDm-enabled device. For example, hardware

13

device **206A** may be a graphics processor, hardware device **206B** may be a network controller, hardware device **206C** may be a storage controller, and **206N** may be a USB expansion card. The IHS **500** is also configured with a virtualized environment including a workspace orchestration service (hypervisor) **504** that can instantiate multiple workspaces (e.g., virtual machines (VMs)) **506 A-N** (or individually referred to as VM **506**). As shown in the present embodiment, workspace orchestration service **504** has configured workspace **506N** to have a fixed pass-through configuration **510** with the graphics controller **206A**. While the present embodiment describes a fixed pass-through configuration **510** that is configured on a graphics processor hardware device **206A**, it should be understood that the teachings of the present disclosure may be applied to other hardware devices **206 B-N** without departing from the spirit and scope of the present disclosure.

The BMC **132** includes executable code for continual monitoring one or more operating characteristics of any hardware devices **206 A-N** that are operating in a fixed pass-through configuration **510** with a workspace **506 A-N** such that when it determines that the operating characteristics are indicative of a security breach of the fixed pass-through configuration **510**, it may perform one or more operations to quarantine the one hardware device **206 A-N**. The operating characteristics may be any type that may be indicative of a security breach of the fixed pass-through configuration **510**. Examples of such operational characteristics may include input/output I/O device faults, correctable errors, uncorrectable errors, improper memory access requests, and the like.

Quarantining the hardware device **206A** may be useful to ensure that security breaches existing on the hardware device **206A** are not propagated to other hardware devices **206 A-N** in the SPDM-based trusted network **502**. For example, once the graphics processor hardware device **206A** is compromised, it may be loaded with malicious code that can be executed to corrupt the NID hardware device **206B**, which can in turn, be loaded with additional malicious code to corrupt the storage controller hardware device **206C** and so on. Thus, by quarantining the hardware device **206** that is affected by the corrupted fixed pass-through configuration **510**, other hardware devices **206** in the IHS **100** may be secured from any malicious security breaches.

The BMC **132** may perform any suitable type of operation to quarantine the affected hardware device **206A**. In one embodiment, the I/O operations of the hardware device **206** may be disabled (e.g., removed), such as described above with reference to step **356** of FIG. **3** or step **454** of FIG. **4**. In another embodiment, the BMC **132** may communicate with the BMC software agent **202**, that in turn, communicates with the workspace orchestration system **504** (e.g., via a vmkdevmgr device) to initiate a device detach operation from PCIe layer. In yet another embodiment, the BMC **132** may communicate with the BMC software agent **202**, that in turn, communicates with the ACPI layer of the IHS **500** to initiate an ACPI_OSC method to turn off the hardware device **206** by setting its appropriate ACPI device state (e.g., D3 cold).

The workspace orchestration system **504**, which may also be referred to as a hypervisor or virtual machine manager (VMM), is capable of instantiating multiple instances of workspaces **506a-n** (e.g., guest operating systems (OSs)) on the IHS **500**. Many currently available workspace orchestration systems **502** provide a fixed pass-through feature. This fixed pass-through feature, however, causes certain security vulnerabilities. One example of such a fixed pass-

14

through feature is a PCIe pass-through that may be utilized with PCIe system buses. While assigning a hardware device to be directly controlled by a VM improves its performance, this approach also introduces a wide range of security vulnerabilities such as described above. Accordingly, the BMC **132** is configured to quarantine hardware devices **206** with fixed passthroughs that exhibit security breaches so that the rest of the hardware devices **206** may be protected according to embodiments of the present disclosure.

FIG. **6** illustrates an example fixed pass-through monitoring method **600** for hardware devices of an information handling system (IHS) using a baseboard management controller (BMC) according to one embodiment of the present disclosure. As will be described in detail herein below, the method **600** may be provided for continual monitoring of one or more operating characteristics of a hardware device **206** with a fixed pass-through configuration such that, when the operational characteristics exhibit a security breach, the hardware device **206** may be quarantined to protect the rest of the hardware devices **206** in the IHS **100**. The method **600** may be performed in whole, or at least in part by either the BMC **132**, the BMC software agent **202**, or a combination thereof. Initially, the IHS **100** is configured with multiple SPDM enabled hardware devices **206 A-N**.

At step **602**, the IHS is powered on. Thereafter at step **604**, the BMC **132** establishes a SPDM-based trusted network **502** with the hardware devices **206 a-n**. For example, the steps as described above with reference to FIG. **3** or **4** may be performed with each hardware device **206** so that it may be added to the SPDM-based trusted network **502**. The workspace orchestration system **504** then instantiates a workspace **506** that has a fixed pass-through configuration **510**. In one embodiment, the fixed pass-through configuration **510** comprises a PCIe fixed pass-through configuration at step **606**.

While the workspace **506** is being executed, the method **600** continually monitors certain operating characteristics of fixed pass-through configuration at step **608**. The method **600** may monitor any operational characteristics that may be indicative of a security breach through the fixed pass-through configuration **510**. Additional details of certain operational characteristics that may be monitored by the BMC **132** and/or BMC software agent **202** will be described in detail herein below.

At step **610**, The method **600** determines that a security breach is detected. In one embodiment, the method **600** may make the determination based on one or more policies. For example, the method **600** may determine that a security breach exists when a certain operational characteristics event occurs once. As another example, the method **600** may determine that a security breach exists when a certain operational characteristics event occurs a specified quantity of times over a certain elapsed period of time. For yet another example, the method **600** may determine that a security breach exists when a certain pattern of operational characteristics events occur, such as a certain sequence of operational characteristics events. For yet another example, the method **600** may determine that a security breach exists when a machine learning (ML) process indicates that a certain pattern of operational characteristics events may be indicative of a security breach. Nevertheless, if the method **600** determines whether or not a security breach is detected, processing continues at step **612**; otherwise, processing continues at step **608** to continually monitor the operational characteristics of the fixed pass-through configuration **510**.

At step **612**, the method **600** analyzes the affected hardware device **206**. For example, the BMC software agent **202**

15

may analyze a memory dump of the affected hardware device **206** to identify any specific root causes of the security breach, and transmit information associated with the analysis to the BMC **132**, which in turn, may send the information to an administrator computer in the form of an asynchronous event notification (AEN).

At step **614**, the method **600** quarantines the hardware device **206**. Quarantining the hardware device **206** effectively protects the other hardware devices **206** from being adversely affected by the security breach. The hardware device **206** may be quarantined in any suitable manner. Embodiments of various techniques for quarantining the hardware device **206** are described herein above with respect to FIG. **5** and below with respect to FIGS. **7** and **8**.

At this point, the affected hardware device **206** is quarantined and is inhibited from interacting with the other hardware devices **206** in the IHS **100**. Nevertheless, steps **616** through **620** describe one embodiment of how the affected hardware device **206** may be processed so that its quarantine state can be alleviated (e.g., the affected hardware device **206** can be returned to normal operating conditions).

At step **616**, the method **600** checks for any available firmware or security patch updates for the affected hardware device **206**. For example, the method **600** may access an online support portal for the hardware device **206** to determine whether any firmware updates are available for the affected hardware device **206** and if so, download the firmware update(s), and flash the affected hardware device **206** with the downloaded firmware update(s). After the firmware update(s) have been flashed, the hardware device **206** may be re-booted to allow the firmware update(s) to become effective.

At step **618**, the method **600** determines whether or not the firmware has been updated on the affected hardware device **206**. If not, processing continues at step **616** to again check for any firmware updates that may be available for the hardware device **206**. If, however, the firmware has been updated, processing continues at step **620** to re-establish the SPDm-based trust with hardware device **206**. For example, the steps as described above with reference to FIG. **3** or **4** may be performed with the affected hardware device **206** so that it may be added to the SPDm-based trusted network **502**. When the hardware device **206** has been added to the SPDm-based trusted network **502**, processing may continue at step **608** for continual monitoring of the fixed pass-through configuration **510** of the hardware device **206** as well as the fixed pass-through configuration **510** of other hardware devices **206** in the IHS **100**.

The method **600** described above may be continually performed for monitoring the fixed pass-through configuration **510** of a workspace **506** as long as that workspace **506** is being executed on the IHS **100**. Nevertheless, when use of the workspace migration method **300** is no longer needed or desired, the method **600** ends.

Although FIG. **6** describes an example method that may be performed for monitoring and remediating the fixed pass-through configuration **510** of a workspace **506**, the features of the method **600** may be embodied in other specific forms without deviating from the spirit and scope of the present disclosure. For example, the method **600** may perform additional, fewer, or different operations than those described in the present examples. As another example, certain steps of the method **600** may be performed by other components in the IHS **100** other than those described above.

FIG. **7** is a workflow diagram describing certain steps of another fixed pass-through monitoring and security breach

16

remediation method **700** using a software driver associated with the processor according to one embodiment of the present disclosure. Initially, the IHS **100** is configured with a BMC **132**, a hardware device **206a** that can communicate using the SPDm protocol, and a fixed pass-through driver **730**, such as one provided for use with processor devices having a VT-d fixed pass-through feature provided by INTEL or an I/O management unit (IOMMU) fixed pass-through feature provided by AMD. Although only one hardware device **206** is shown and described, it should be understood that the method **700** can be practiced with any quantity of hardware devices **206**, such as two or more SPDm enabled hardware devices **206**.

At step **702**, the BMC software agent **202** communicates with the fixed pass-through driver **730** to setup a subscription to certain operational characteristics detected by the processor associated with the fixed pass-through driver **730**. For example, the BMC software agent **202** may subscribe to receive an event notification when the fixed pass-through driver **730** detects a page fault event or other suitable operational characteristics exhibited by the processor. At this point, the processor is being continually monitored to detect an operational characteristics event that may be indicative of a security breach of the fixed pass-through configuration **510**. Although the present embodiment describes a page fault event as an operational characteristic that is used to determine that a security breach has occurred, it should be understood that other operational characteristics exist that may be used to determine that a security breach has occurred without departing from the spirit and scope of the present disclosure.

At step **704**, the fixed pass-through driver **730** detects that a page fault event has occurred. As such, the fixed pass-through driver **730** sends an event notification to the BMC software agent **202** in accordance with one or more rules of the subscription established at step **702**. The BMC software agent **202** then forwards the event notification to the BMC **132** at step **708**. In response, the BMC **132** disables the trust (quarantines) of the hardware device **206** (e.g., removes the hardware device **206** from the SPDm-based trusted network) at step **710**. For example, the I/O operations of the hardware device **206** may be disabled, such as described above with reference to step **356** of FIG. **3** or step **454** of FIG. **4**. It may be important to note that the fixed pass-through driver **730** is configured to notify the BMC software agent **202** rather than what is conventionally performed by invoking an assert flag on the workspace orchestration system **504** and/or the operating system of the IHS **500**. Additionally, the BMC software agent **202** further quarantines the hardware device **206** by detaching operation of the hardware device **206** from the system at step **712**. For example, the BMC software agent **202** may invoke a PCIe 'disable ioctl' command or a PCIe device driver detach operation against the specific device to stop its I/O and detach the device from the device layer.

At step **714**, the BMC software agent **202** may quarantine the hardware device **206** by sending a notification message to the ACPI layer **735** of the IHS **500**, which causes the ACPI layer **735** to turn the hardware device **206** off at step **716**. The BMC software agent **202** may also quarantine the hardware device **206** by triggering an API to de-allocate the memory address range used by the hardware device **206** at step **718**. Such a procedure may be at least somewhat similar to a memory page retire feature that is used to ensure that the memory page is not used by other processes in the IHS **500**.

FIG. **8** is a workflow diagram describing certain steps of yet another fixed pass-through monitoring and security

17

breach remediation method **800** by monitoring uncorrectable errors of the fixed pass-through configuration according to one embodiment of the present disclosure. Initially, the IHS **100** is configured with a BMC **132**, a hardware device **206a** that is capable of communicating using the SPDMM 5 protocol, and a fixed pass-through driver **830**, such as one provided for use with processor devices having a VT-d fixed pass-through feature provided by INTEL or an I/O management unit (IOMMU) fixed pass-through feature provided by AMD. Although only one hardware device **206** is shown and described, it should be understood that the method **800** can be practiced with any quantity of hardware devices **206**, such as two or more SPDMM enabled hardware devices **206**.

At step **802**, a communication channel is established between the workspace orchestration system **504** and a PCIe device driver **830** associated with the workspace **506**. In one embodiment, the communication channel may be established using a virtual machine communication interface (VMCI) provided by VMWARE. At step **804**, the BMC software agent **202** communicates with the workspace orchestration system **504** to setup a subscription to certain operational characteristics detected by the PCIe device driver **830**. In one embodiment, the BMC software agent **202** may setup a subscription to a device manager (e.g., vmkdevmgr provided by VMWARE) using a custom kernel ioctl command or using a vmkernel callback API. Additionally, the device manager of the workspace orchestration system **504** may be subscribed to a PCI Advanced Error Reporting (AER) root driver. At this point, a communication channel is established between PCIe AER root driver and the BMC **132** via the BMC software agent **202** and device manager of the workspace orchestration system **504** such that the BMC **132** may receive and process uncorrectable errors generated by the fixed pass-through configuration **510** of the workspace **506**.

At step **806**, the workspace orchestration system **504**, via the PCI AER root driver determines that an uncorrectable error (e.g., operational characteristics event) has occurred with the fixed pass-through configuration **510** of hardware device **206**. In one embodiment, the device manager of the workspace orchestration system **504** possesses the ability to know whether the operational characteristics event is a correctable or an uncorrectable error using AER's severity status bit. When this occurs, the workspace orchestration system **504** sends this information along with identity information associated with the hardware device **206** to the BMC software agent **202** at step **808**. In one embodiment, the workspace orchestration system **504** may send PCI block/device/function (BDF) information that uniquely identifies the affected hardware device **206** on the PCIe system bus.

At step **810**, the BMC software agent **202** forwards this information to the BMC **132**, which determines whether the operational characteristics event is indicative of a security breach at step **812**. In one embodiment, the BMC **132** determines that if the AER portion of the workspace orchestration system **504** reports an error and if the BMC **132** SEL is not reported with a valid PCIe fatal error, the operational characteristics event will be treated as a security breach. In such a case, the BMC **132** disables the trust (quarantines) of the hardware device **206** at step **814**. For example, the I/O operations of the hardware device **206** may be disabled, such as described above with reference to step **356** of FIG. **3** or step **454** of FIG. **4**.

Additionally, the BMC software agent **202** may further quarantine the hardware device **206** by performing a detach operation of the hardware device **206** from the system at steps **816**, **818**, and **820**. For example, the BMC **132**, via the

18

BMC software agent **202**, may communicate with the device manager (e.g., vmkdevmgr) of the workspace orchestration system **504** to initiate a PCIe device driver detach operation against the specific hardware device **206** to stop its I/O and detach the device from the device layer. At step **822**, the BMC **132** may communicate with the BMC software agent **202**, that in turn, communicates with the ACPI layer of the IHS **500** to initiate an ACPI_OSC method to turn off the hardware device **206** by setting its ACPI device state (e.g., D3 cold) at step **824**.

Although the methods **700** and **800** of FIGS. **7** and **8**, respectively, each describe an example method that may be performed for managing event messages, the features of the methods **700** and **800** may be embodied in other specific forms without deviating from the spirit and scope of the present disclosure. For example, either of the methods **700** and **800** may perform additional, fewer, or different operations than those described in the present examples. As another example, certain steps of the methods **700** and **800** may be performed by other components in the IHS **100** other than those described above.

It should be understood that various operations described herein may be implemented in software executed by logic or processing circuitry, hardware, or a combination thereof. The order in which each operation of a given method is performed may be changed, and various operations may be added, reordered, combined, omitted, modified, etc. It is intended that the invention(s) described herein embrace all such modifications and changes and, accordingly, the above description should be regarded in an illustrative rather than a restrictive sense.

Although the invention(s) is/are described herein with reference to specific embodiments, various modifications and changes can be made without departing from the scope of the present invention(s), as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present invention(s). Any benefits, advantages, or solutions to problems that are described herein with regard to specific embodiments are not intended to be construed as a critical, required, or essential feature or element of any or all the claims.

Unless stated otherwise, terms such as "first" and "second" are used to arbitrarily distinguish between the elements such terms describe. Thus, these terms are not necessarily intended to indicate temporal or other prioritization of such elements. The terms "coupled" or "operably coupled" are defined as connected, although not necessarily directly, and not necessarily mechanically. The terms "a" and "an" are defined as one or more unless stated otherwise. The terms "comprise" (and any form of comprise, such as "comprises" and "comprising"), "have" (and any form of have, such as "has" and "having"), "include" (and any form of include, such as "includes" and "including") and "contain" (and any form of contain, such as "contains" and "containing") are open-ended linking verbs. As a result, a system, device, or apparatus that "comprises," "has," "includes" or "contains" one or more elements possesses those one or more elements but is not limited to possessing only those one or more elements. Similarly, a method or process that "comprises," "has," "includes" or "contains" one or more operations possesses those one or more operations but is not limited to possessing only those one or more operations.

The invention claimed is:

1. An Information Handling System (IHS), comprising: a plurality of hardware devices; and

a Baseboard Management Controller (BMC) in communication with the plurality of hardware devices, the BMC comprising instructions that are executable by at least one processor to:

monitor one or more operating characteristics of at least one hardware device of the plurality of hardware devices to determine whether or not a fault comprising the one or more operating characteristics has been occurring at a rate that exceeds a specified threshold, wherein the at least one hardware device is operating in a fixed pass-through configuration with a workspace, wherein the workspace has been instantiated by a workspace orchestration service executed on the IHS, and wherein the one or more operating characteristics comprise at least one of: an input/output (I/O) device fault, a correctable error, an uncorrectable error, an improper memory access request, or a page fault, and wherein the at least one hardware device comprises a Security Protocol and Data Model (SPDM)-enabled hardware device;

perform a mutual authentication procedure with the SPDM-enabled hardware device to form a SPDM-based trusted network between the SPDM-enabled hardware device and the BMC;

determine that the operating characteristics are indicative of a security breach of the fixed pass-through configuration; and

perform an operation to quarantine the at least one SPDM-enabled hardware device when the fixed pass-through configuration is determined to possess the security breach by maintaining the quarantined at least one SPDM-enabled hardware device in a quarantine state until a firmware update procedure has been performed on the at least one SPDM-enabled hardware device.

2. The IHS of claim 1, wherein the instructions are further executed to perform the operation to quarantine the at least one hardware device by removing the SPDM-enabled hardware device from the SPDM-based trusted network.

3. The IHS of claim 1, wherein the instructions are further executed to perform the operation to quarantine the at least one hardware device by performing a device detach operation on the SPDM-enabled hardware device.

4. The IHS of claim 1, wherein the instructions are further executed to perform the operation to quarantine the at least one hardware device by setting an Advanced Configuration and Power Interface (ACPI) state of the SPDM-enabled hardware device to an off state.

5. The IHS of claim 1, wherein the fixed pass-through configuration comprises a Peripheral Component Interconnect Express (PCIe) pass-through configuration.

6. The IHS of claim 1, wherein the fault comprises at least one of an input/output memory management unit (IOMMU) page fault, a VT-d page fault, or a Peripheral Component Interconnect Express (PCIe) uncorrectable error.

7. A fixed pass-through monitoring method comprising: monitoring, using instructions stored in at least one memory and executed by at least one processor, one or more operating characteristics of at least one hardware device of a plurality of hardware devices to determine whether or not a fault comprising the one or more operating characteristics has been occurring at a rate that exceeds a specified threshold, wherein the at least one hardware device is operating in a fixed pass-through configuration with a workspace using a Baseboard Management Controller (BMC) in communication with the plurality of hardware devices of an

Information Handling System (IHS), wherein the workspace has been instantiated by a workspace orchestration service executed on the IHS, wherein the one or more operating characteristics comprise at least one of: an input/output (I/O) device fault, an improper memory access request, or a page fault, and wherein the at least one hardware device comprises a Security Protocol and Data Model (SPDM)-enabled hardware device;

performing a mutual authentication procedure with the SPDM-enabled hardware device to form a SPDM-based trusted network between the SPDM-enabled hardware device and the BMC;

determining, using the instructions, that the one or more operating characteristics are indicative of a security breach of the fixed pass-through configuration, based at least in part on determining that at least one of the one or more operating characteristics represents an uncorrectable error; and

performing, using the instructions, an operation to quarantine the at least one SPDM-enabled hardware device when the fixed pass-through configuration is determined to possess the security breach by maintaining the quarantined at least one SPDM-enabled hardware device in a quarantine state until a firmware update procedure has been performed on the at least one SPDM-enabled hardware device.

8. The fixed pass-through monitoring method of claim 7, further comprising performing the operation to quarantine the at least one hardware device by removing the SPDM-enabled hardware device from the SPDM-based trusted network.

9. The fixed pass-through monitoring method of claim 7, further comprising performing the operation to quarantine the at least one hardware device by performing a device detach operation on the SPDM-enabled hardware device.

10. The fixed pass-through monitoring method of claim 7, further comprising performing the operation to quarantine the at least one hardware device by setting an Advanced Configuration and Power Interface (ACPI) state of the SPDM-enabled hardware device to an off state.

11. An Information Handling System (IHS), comprising: first and second Security Protocol and Data Model (SPDM)-enabled hardware devices, the first SPDM-enabled hardware device comprising instructions that are executable by at least one processor to:

monitor one or more operating characteristics of the second SPDM-enabled hardware device, to determine whether or not a fault comprising one or more operating characteristics has been occurring at a rate that exceeds a specified threshold, wherein the second SPDM-enabled hardware device is operating in a fixed pass-through configuration with a workspace, wherein the workspace has been instantiated by a workspace orchestration service executed on the IHS, and wherein the one or more operating characteristics comprise at least one of: an input/output (I/O) device fault, an improper memory access request, or a page fault;

perform a mutual authentication procedure with the second SPDM-enabled hardware device to form a SPDM-based trusted network between the second SPDM-enabled hardware device and the BMC;

determine that the operating characteristics are indicative of a security breach of the fixed pass-through configuration, wherein the determination is based at least in part on determining that at least one of the one or more operating characteristics represents an uncorrectable

21

error, using error severity information obtained from an interface bus configured to be coupled with the second hardware device; and

perform an operation to quarantine the second SPDME-enabled hardware device when the fixed pass-through configuration is determined to possess the security breach by maintaining the quarantined second SPDME-enabled hardware device in a quarantine state until a firmware update procedure has been performed on the second SPDME-enabled hardware device.

12. The IHS of claim 11, wherein the instructions are further executed to perform the operation to quarantine the second SPDME-enabled hardware device by at least one of removing the second SPDME-enabled hardware device from the SPDME-based trusted network, performing a device detach operation on the second SPDME-enabled hardware device, or by setting an Advanced Configuration and Power Interface (ACPI) state of the second SPDME-enabled hardware device to an off state.

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20

22